7.1 For linear polarization,

$S_0 = a_1^2 + a_2^2$, $S_1 = a_1^2 - a_2^2$

$S_2 = 2a_1a_2\cos(\delta_2 - \delta_1)$,

$S_3 = 2a_1a_2\sin(\delta_2 - \delta_1)$.

then,

$$a_1 = \left(\frac{S_0 + S_1}{2}\right)^{1/2},$$

$$a_2 = \left(\frac{S_0 - S_1}{2}\right)^{1/2},$$

$$\delta_2 - \delta_1 = \arctan\left(\frac{S_3}{S_2}\right).$$

For circular polarization,

$S_0 = a_+^2 + a_-^2$,

$S_1 = 2a_+a_-\cos(\delta_- - \delta_+)$

$S_2 = 2a_+a_-\sin(\delta_- - \delta_+)$

$S_3 = a_+^2 - a_-^2$.

then,

$$a_+ = \left(\frac{S_0 + S_3}{2}\right)^{1/2}$$

$$a_- = \left(\frac{S_0 - S_3}{2}\right)^{1/2}$$

$$\delta_- - \delta_+ = \arctan\left(\frac{S_2}{S_1}\right)$$

(a) Linear polarization,

$a_1 = 1$, $a_2 = \sqrt{2}$

$\delta_2 - \delta_1 = -\frac{\pi}{4}$,

Circular polarization.

$a_+ = \frac{\sqrt{2}}{2}$, $a_- = \frac{\sqrt{10}}{2}$

$\delta_- - \delta_+ = -\arctan 2$

(b) Linear polarization.

$a_1 = \frac{5\sqrt{2}}{2}$, $a_2 = \frac{5\sqrt{2}}{2}$.

$\delta_2 - \delta_1 = \arctan\left(\frac{7}{24}\right)$.

Circular polarization

$a_+ = 4$, $a_- = 3$

$\delta_- - \delta_+ = \frac{\pi}{2}$.